Effects of the Junction Temperature on the Dynamic Resistance of White LEDs

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***Abstract*—This paper deals with the thermal characteristics of the I–V curve of GaN-based white light-emitting diodes (LEDs), focused on the variations of the dynamic resistance. The final goal of this study is to improve the static and dynamic operations of the LED driver within a wide range of temperatures. Four LEDs from different manufacturers were chosen for this study. The first part of this paper shows the thermal characterization of the forward voltage at a given injected current. After that, the experimental data are fitted in order to calculate the junction temperature accurately. Then, a small-signal analysis where the LEDs are supplied with dc current and an ac perturbation superimposed at the operation point under variable junction temperature is covered. This analysis allows the dynamic resistance to be experimentally determined for a wide junction temperature range. Furthermore, the experimental data have been fitted in order to establish the relationship between junction temperature and dynamic-resistance variation, so the dynamic resistance can be determined for a given operation point. Finally, an illustrative example is presented as a case study in order to analyze the implications of the dynamic resistance on the output-current ripple and on the closed-loop operation of a LED driver. The experimental results confirm that the junction temperature shift induces a variation in the dynamic resistance, which might have a significant effect on the output-current ripple and closed-loop performance in certain LED fixtures.**

***Index Terms*—Closed-loop systems, control design, current–voltage characteristics, electrical ballasts, impedance measurement, LED lamps, light-emitting diodes, lighting, optoelectronic devices, power electronics, semiconductor devices, thermal analysis, threshold voltage, voltage measurement, III-V semiconductor materials.**

Manuscript received February 24, 2012; revised May 21, 2012; accepted June 1, 2012. Date of publication January 25, 2013; date of current version March 15, 2013. Paper 2012-ILDC-094.R1, presented at the 2012 IEEE Applied Power Electronics Conference and Exposition, Orlando, FL, USA, February 5–9, and approved for publication in the IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS by the Industrial Lighting and Display Committee of the IEEE Industry Applications Society.

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Digital Object Identifier 10.1109/TIA.2013.2243092

I. INTRODUCTION

LIGHT-EMITTING diodes (LEDs) have become an outstanding technology in lighting due to their extremely long life span—several thousands of hours for a lumen maintenance of 70%, i.e., L70 [1], [2], under the Illuminating Engineering Society LM-80-08 procedure [3]—and their continually increasing luminous efficacy, which has recently overcome that of compact fluorescent lamps and metal-halide lamps. There is a growing trend toward replacing the traditional light sources with LED retrofits and extending their use in most general applications, such as street lighting, outdoor/indoor general lighting, architectural or ambiance lighting, and automotive lamps. However, their electrical characteristics make the driver implementation demanding. In this way, many works have been developed in order to achieve a deeper insight into their behavior as an electric power load [4]–[6], whereas other works have linked their light and electrical characteristics globally as a nonlinear lighting device [7]–[9]. Due to the nonlinear electrical behavior of LEDs, it is very important to consider an accurate model in order to design a LED driver properly. The simplest choice is the equivalent resistor at the operation point. However, this approximation is valid only for the operation point, leading to errors in a small-signal analysis [10]. The most usually employed model in designing LED drivers is the one consisting of an ideal diode, a voltage source, and a resistor in series, where the voltage source and the resistor stand for the threshold voltage and the dynamic resistance, respectively [5], [6]. Moreover, as the junction temperature affects their I–V curve, lowering the voltage drop for a given injected current as the temperature rises, it is widely assumed that the best control technique for power LEDs relies on a current-controlled scheme [11].

As a result, the operation condition of closed-loop converters will lead to small variations around the operation point. Moreover, the forward voltage variation affects the dc gain of the duty ratio to LED-current transfer function, and in the case of buck–boost- and boost-based converters, the right-half-plane zero is also affected [10]. Manufacturers usually provide the temperature coefficient for the forward voltage related to a junction temperature of 25 °C, so the expected forward voltage V_D can be calculated using

$$V_D = V_D(25\ ^\circ\text{C}) + \lambda\,(T_j(^\circ\text{C}) - 25) \tag{1}$$

where λ is the temperature coefficient of forward voltage, i.e., $\partial V_D/\partial T_j$. Nevertheless, there are few studies about the thermal effects on the dynamic resistance, such as [12]. These variations could likely have a great impact on the closed-loop dynamics,

TABLE I
MAIN TYPICAL CHARACTERISTICS OF THE LEDs UNDER TEST

LED	T_j (max.) (ºC)	I_D (max.) (A)	$R_{\theta j\text{-}c}$(ºC/W)	λ (mV/ºC)
Dragon+	125	1.0	11 (max.)	Chart [13]
K2	150	1.5	5.5	-2.8
P4	145 (0.7A)	1.0 (90ºC)	6.9	Chart [15]
XR-E	150	1.0	8	-4

since the slight changes for single LED dynamic resistance would be adding up together in large strings of series-connected devices. Therefore, the need for a better understanding of LED small-signal behavior and its junction temperature dependence arises.

This paper will deal with the dynamic-resistance change related to the junction temperature of four commercial devices: Osram Golden Dragon Plus [13], Lumileds Luxeon K2 with thin-film flip chip (TFFC) [14], Seoul Semiconductors Z-Power P4 [15], and Cree XLamp XR-E [16]. Their main characteristics are gathered in Table I, where T_j is the maximum junction temperature, I_D is the maximum forward current, $R_{\theta j-c}$ is the thermal resistance between the junction and case, and λ is the temperature coefficient of forward voltage, as extracted from the datasheets [14], [16].

Section II will introduce the procedure for determining the I–V characteristic for a wide range of junction temperatures. The experimental results obtained will also be shown in this section. Section III will introduce the setup for the small-signal analysis for variable junction temperature as well as the experimental results obtained. Section IV will discuss the implications of the dynamic-resistance shift in the performance of LED lighting fixtures. Finally, Section V will summarize the conclusions achieved in this work.

II. FORWARD VOLTAGE CHARACTERIZATION PROCEDURE

It has extensively been reported that the most accurate procedure for determining the junction temperature at a given injected current is the method consisting of a pulsed operation of the LED in a temperature-controlled chamber [17]–[20]. In this method, the device under test (DUT) is placed in a temperature-controlled room and tested under pulsed-current injection with a very small duty cycle. It is assumed that, if the duty cycle is small enough, typically 0.1%, the self-heating of the p-n junction is negligible [17], and therefore, the junction temperature will be the same as the board and heat-sink temperatures, which, in turn, are the same as the ambient temperature. Therefore, after measuring the forward voltage drop for each injection current at each ambient temperature, the V_D–T_j characteristic of the DUT can be established for each forward current with an accuracy around ± 3 °C [19].

One sample of each LED was fixed to a metal-core printed circuit board (MCPCB) with thermal bond. A K-type thermocouple was attached to the MCPCB in order to measure the board temperature. Then, the entire fixture was placed inside a temperature-controlled chamber. By means of a specific driver, a 20-μs-width current pulse was injected to the DUT for each

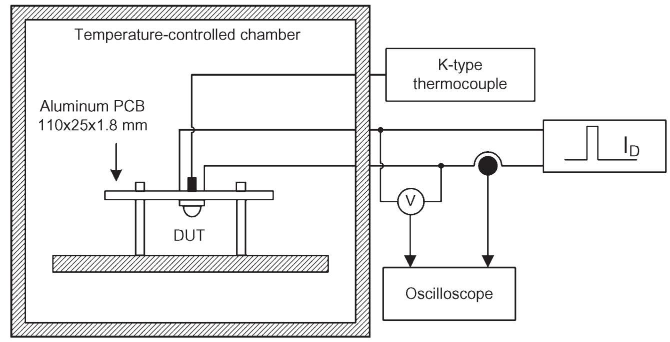

Fig. 1. Experimental arrangement in order to establish the forward-voltage-versus-junction-temperature relation for a given DUT.

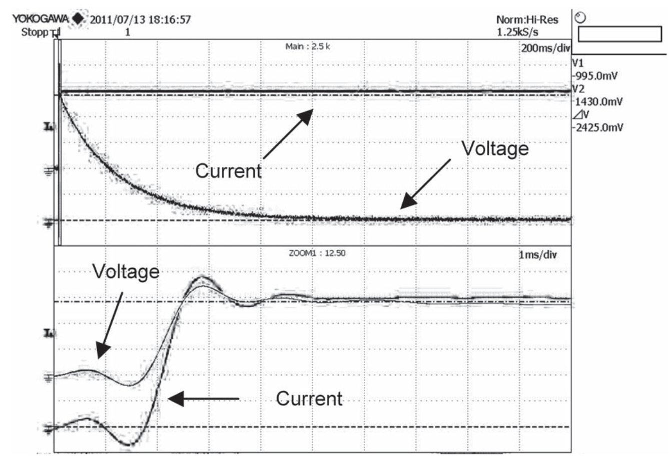

Fig. 2. One-thousand-milliampere current pulse test to determine the forward voltage drop of the Cree XR-E XLamp device. (Upper) Forward current and forward voltage of the DUT. Vertical scale: 200 mA/div and 500 mV/div. Horizontal scale: 200 ms/div. (Lower) Zoom of the pulse rising edge. Vertical scale: 200 mA/div and 500 mV/div. Horizontal scale: 1 ms/div. Channel 1: dc coupling. Channel 2: ac coupling.

target current level at each target temperature. The experimental arrangement is shown in Fig. 1. The pulsewidth was chosen as a tradeoff between high SNR and negligible LED self-heating, provided that the pulsewidth is much lower than the thermal time constant of the LEDs. This thermal time constant is considered to be in the millisecond range for the chip [4], [22]. In order to check whether a 20-μs-width current pulse induces a nonnegligible self-heating to the DUT, a 1000-mA current step was applied to a Cree X-RE XLamp device, measuring the forward voltage drop for a long time extent. Fig. 2 shows the results of the test, from which it can be assumed that the forward voltage drop of the DUT can be considered negligible for a pulsewidth of less than 20 μs.

The devices were tested from 100 mA up to 1 A in 100-mA steps, plus the nominal current, i.e., 350 mA, for an ambient temperature ranging from -20 °C to 120 °C in 10 °C steps, plus 25 °C. Moreover, this experimental procedure allows the temperature coefficient of forward voltage λ to be obtained.

Figs. 3–6 show the experimental data along with the second-order polynomial fit obtained for the Golden Dragon Plus, Luxeon K2, Z-Power P4, and XLamp XR-E devices, respectively.

Once this procedure was developed, some differences rose up: Whereas the Golden Dragon and Luxeon K2 devices showed a rather linear dependence of the forward voltage over

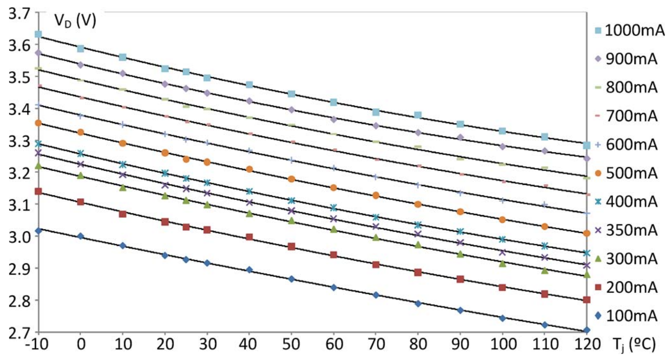

Fig. 3. Experimental results for the dependence of the forward voltage as a function of the junction temperature for the Golden Dragon Plus device.

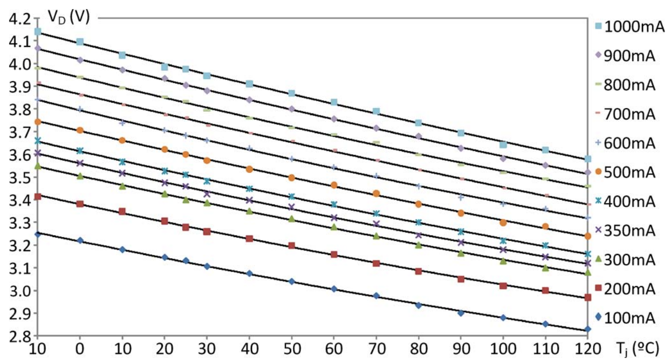

Fig. 4. Experimental results for the dependence of the forward voltage as a function of the junction temperature for the Luxeon K2 with TFFC device.

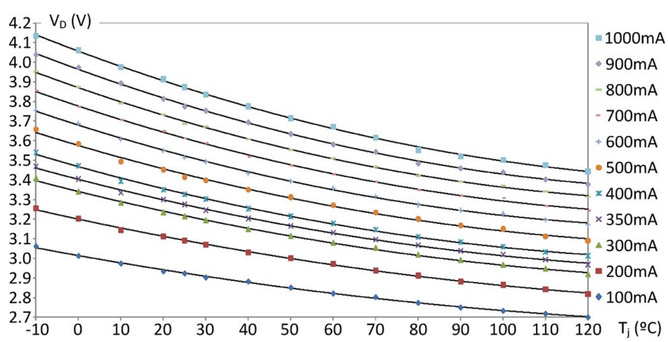

Fig. 5. Experimental results for the dependence of the forward voltage as a function of the junction temperature for the Z-Power P4 device.

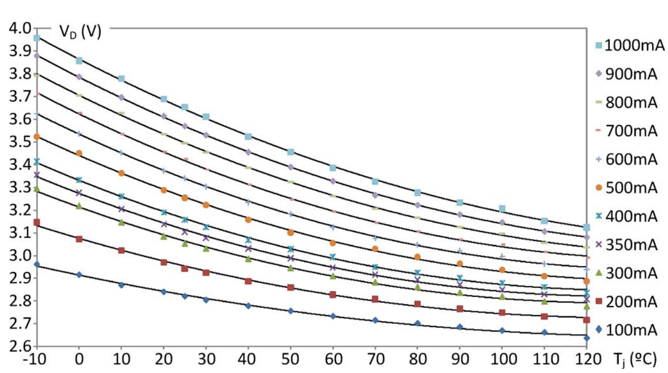

Fig. 6. Experimental results for the dependence of the forward voltage as a function of the junction temperature for the XLamp XR-E device.

junction temperature, XLamp XR-E and Z-Power P4 showed a pronounced quadratic dependence. This effect may be caused by the different internal structures of each device.

TABLE II
EXPERIMENTAL TEMPERATURE COEFFICIENT OF FORWARD VOLTAGE (LINEAR FIT)

LED	λ' (mV/ºC)	Fitting Interval	R^2	λ (mV/ºC)
Golden Dragon Plus	-2.65	@ 350mA -10ºC ≤ Tj ≤ 120 ºC	0.9964	Chart [13]
Luxeon K2 TFFC	-4.30	@1000mA -10ºC ≤ Tj ≤ 120 ºC	0.9976	-2.8
Z.Power P4	-3.41	@350mA 25ºC ≤ Tj ≤ 100 C	0.9877	Chart [15]
X-Lamp XR-E	-3.40	@350mA 25ºC ≤ Tj ≤ 100 C	0.9749	-4

Tables IV–VII show the second-order polynomial fit of the forward voltage drop related to junction temperature for each target current. These tables correspond to the Golden Dragon Plus, Luxeon K2, Z-Power P4, and XLamp XR-E devices, respectively. The correlation coefficient R^2 is also shown. A good fit was achieved, with correlation coefficients around 0.999 and always above 0.995. These polynomials allow the calculation of the forward voltage for a given junction temperature, and inversely, the junction temperature can be estimated after measuring the forward voltage for a given injected current.

Finally, a linear fit following (1) was also developed, in order to estimate the junction temperature more easily, and compared with the data provided by the manufacturers. The forward voltage temperature coefficient λ' calculated from the experiments, the value provided by manufacturers λ, the fitting interval, and the correlation coefficient R^2 are gathered in Table II.

III. DYNAMIC-RESISTANCE CHARACTERIZATION PROCEDURE

The effects of the junction temperature on the I–V curve can be theoretically studied by the Shockley equation in order to make a first qualitative prediction on the behavior of the dynamic resistance. Afterward, an experimental procedure for calculating the dynamic resistance for each operation point will be introduced.

The Shockley equation gives the relationship between the voltage applied to a p-n junction and the forward current, i.e., the I–V curve

$$I_D = I_S e^{\frac{q}{\eta k T_j} V_D} \tag{2}$$

where I_S stands for the reverse saturation current, q is the electron charge, η is the ideality factor, k is the Boltzmann constant, and T_j is the temperature measured in kelvins. As it is well known, the I–V curve is affected by the junction temperature since the reverse saturation current is a function of the temperature and so is the thermal voltage: kT_j/q. Deriving (2) with respect to the forward voltage gives the inverse of the dynamic resistance

$$\frac{1}{R_D} = \frac{\partial I_D}{\partial V_D} = \frac{q}{\eta k T_j} I_S e^{\frac{q}{\eta k T_j} V_D} \tag{3}$$

which, as can be seen, is temperature dependent.

However, there are many nonideal and parasitic effects in real diodes. Taking into account the equivalent series and parallel

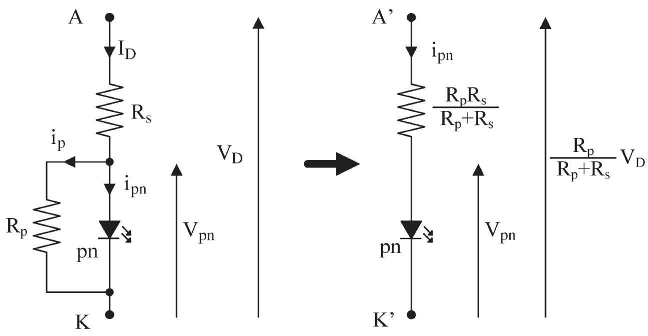

Fig. 7. Nonidealities of a p-n junction: (Left) Series and parallel resistances and (right) its Thevenin equivalent.

resistances of the several layers which LEDs are composed of, the following I–V expression is yielded [17]:

$$I_D = \frac{V_D - R_s I_D}{R_p} + I_S e^{\frac{q}{\eta k T_j}(V_D - I_D R_s)} \tag{4}$$

where R_s and R_p are the series and parallel resistances, respectively. However, this expression induces some difficulties in order to perform further calculations and fit the results by an exponential function. Nevertheless, the nonideal circuit can be simplified by its Thevenin equivalent. These nonidealities, as well as its equivalent circuit, are illustrated in Fig. 7.

The expression for the I–V curve using the Thevenin equivalent, after rearranging and solving for the forward voltage, will be

$$V_D = R_s I_{pn} + K_T \left(1 + \frac{R_s}{R_p}\right) \ln\left(\frac{I_{pn}}{I_S}\right) \tag{5}$$

where K_T stands for $\eta k T_j / q$.

However, the parallel resistance due to leakage current could be negligible in power LEDs [22]. Therefore, under high-forward-current polarization, the effect of the parallel resistance can be neglected, and the current through the p-n junction can be assumed as the forward current.

Then, the dynamic resistance R_D can be calculated as the derivative of forward voltage with respect to forward current

$$R_D = \frac{\partial V_D}{\partial I_D} = R_s + \frac{\eta k T_j}{q} \cdot \frac{1}{I_D} \tag{6}$$

which is indeed affected by the junction temperature T_j and where the ideality factor η is a function of temperature in GaN-based devices [22]. Moreover, the parasitic resistances are likely dependent on temperature as a consequence of the effects of increasing temperature on charge carriers, further activating them in p-type materials [22], and contacts, reducing their parasitic resistance. In addition, the second term in (6) is inversely proportional to forward current. Therefore, a decreasing dynamic resistance with rising forward current will be expected. With regard to the effects of the junction temperature, the behavior of the dynamic resistance will depend on which parameter variation is predominant: either the ideality factor, the series resistance, or the junction temperature.

The dynamic resistance of the four samples was measured using the following procedure, which has previously been introduced in [12]. Each DUT was driven at the three rated current

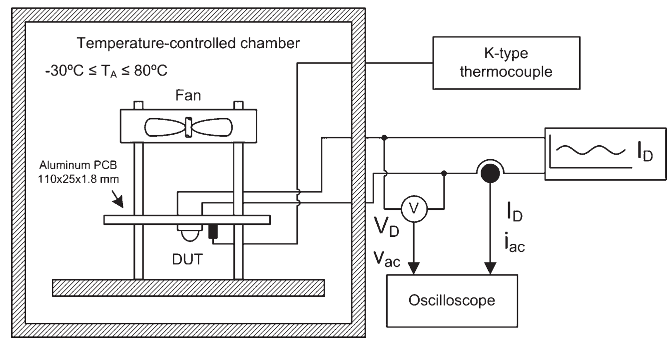

Fig. 8. Experimental arrangement in order to establish the dynamic-resistance-versus-junction-temperature relation for a given DUT.

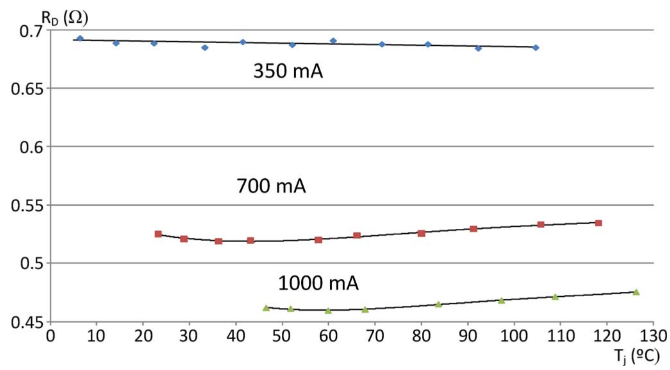

Fig. 9. Experimental results for the dependence of the dynamic resistance as a function of the junction temperature for the Golden Dragon Plus device.

levels I_D: 350, 700, and 1000 mA. A sinusoidal perturbation i_{ac} was superimposed to the dc current using a specific driver. The amplitude of this sinusoidal wave was set to 10% of the dc value and its frequency was set to 1 kHz, in order to avoid thermal effects due to the ac component. Then, the junction temperature of the DUT was modulated in the temperature-controlled chamber with an ambient temperature ranging from -25 °C to 80 °C in 10 °C steps. The junction temperature is determined for each DUT by measuring the forward voltage drop and using the fitting polynomials obtained in Section III. Forced convection was applied in order to reduce the board-to-ambient thermal resistance and the thermal impedance.

Consequently, the dynamic resistance can be calculated as

$$R_D = \left.\frac{v_{ac}}{i_{ac}}\right]_{I_D, T_j} \tag{7}$$

for given forward current and junction temperature I_D and T_j, respectively, where v_{ac} is the ac voltage measured across the LED.

Therefore, the variables to be measured are the ac voltage and ac current, in order to obtain the dynamic resistance; dc current to set the operation point; and dc voltage, in order to determine the junction temperature. This arrangement is shown in Fig. 8.

The results of dynamic resistance as a function of the junction temperature are shown in Figs. 9–12 for the Golden Dragon Plus, Luxeon K2 with TFFC, Z-Power P4, and XLamp XR-E devices, respectively.

With respect to the dynamic resistance related to the forward current, all the devices performed as previously expected, since the dynamic resistance underwent a fall as the injected current

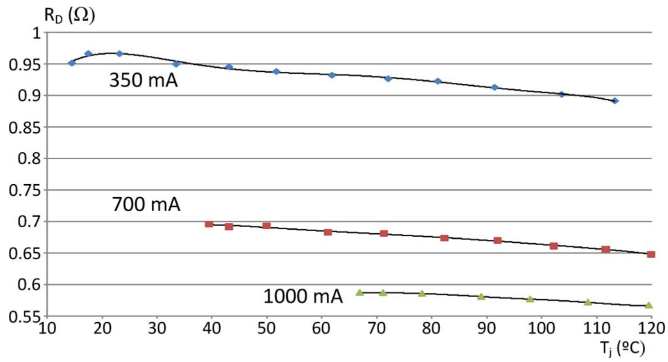

Fig. 10. Experimental results for the dependence of the dynamic resistance as a function of the junction temperature for the Luxeon K2 with TFFC device.

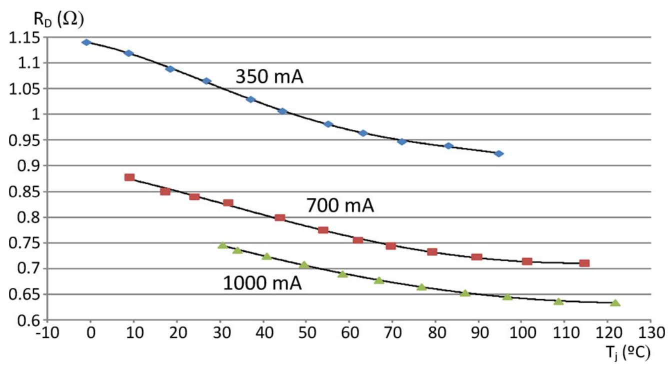

Fig. 11. Experimental results for the dependence of the dynamic resistance as a function of the junction temperature for the Z-Power P4 device.

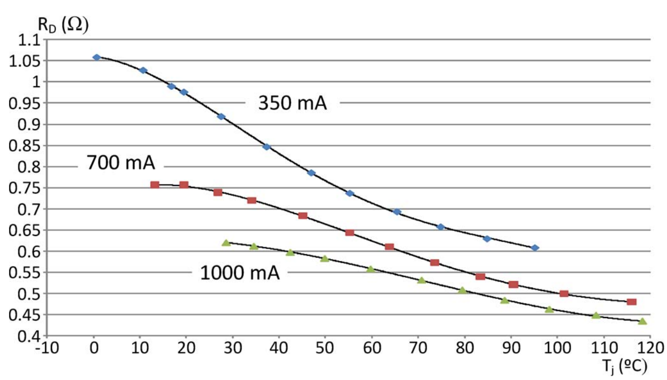

Fig. 12. Experimental results for the dependence of the dynamic resistance as a function of the junction temperature for the XLamp XR-E device.

was stepped up at constant temperature. However, there are slight differences between the four samples under test. Whereas in those devices featuring a more quadratic V_D–T_j dependence, i.e., the XLamp XR-E and the Z-Power P4 devices, the dynamic-resistance values were overlapped for different current levels at different junction temperatures, particularly for the XLamp XR-E device, as can be seen from Figs. 11 and 12; for the devices performing a more linear V_D–T_j relation, i.e., the Luxeon K2 and the Golden Dragon Plus devices, the dynamic-resistance values were clearly delimited for each current level at the whole temperature range, with no dynamic-resistance overlap for the entire junction temperature range.

Regarding the dynamic-resistance shift related to the junction temperature at constant current, the samples performed in different ways, as can be seen from the figures. All the devices except for the Golden Dragon Plus experienced a decrease

TABLE III
EXPERIMENTAL LINEAR MODEL AT 25 °C

LED	350 mA	
	V_γ (V)	R_D (Ω)
Golden Dragon Plus	3.15	0.69
Luxeon K2 TFFC	3.46	0.96
Z-Power P4	3.26	1.07
X-Lamp XR-E	3.11	0.95

on the dynamic resistance as the junction temperature was increased for the three current injection levels: 350, 700, and 1000 mA. In the case of the Golden Dragon Plus device, the dynamic resistance at the 350-mA test remained fairly constant for the entire temperature range, whereas for the 700- and 1000-mA tests, the dynamic resistance featured a decrease at the lowest ambient temperatures, followed by an increase with lowering slope after a minimum corresponding to an ambient temperature of -10 °C, as can be seen from Fig. 9.

In the case of the Luxeon K2 with TFFC device, the dynamic resistance at the 350-mA run showed a maximum at low ambient temperatures, decreasing toward low and high temperatures. At the injection current levels of 700 and 1000 mA, the dynamic resistance seemed to reach a maximum at the lowest ambient temperatures and a decrease with increasing temperature, as can be seen in Fig. 10. Finally and as in the V_D–T_j characterization, the XLamp XR-E and the Z-Power P4 devices experienced a very similar behavior. Both featured a fall in dynamic resistance as the junction temperature rose, with decreasing slope at the lowest and highest temperatures, suggesting maximums and minimums at those levels, respectively.

The highest shift in dynamic resistance corresponds to the XLamp XR-E device at 350 mA, with a drop from 1.06 down to 0.61 Ω, i.e., 42%. On the contrary, the lowest drop in dynamic resistance was featured by the Golden Dragon Plus, with a fairly constant curve at 350 mA.

IV. IMPLICATIONS OF THE DYNAMIC-RESISTANCE SHIFT

With the experimental procedures described in this paper, the actual linear model as

$$V_D|_{T_j} = V_\gamma|_{T_j} + R_D|_{T_j} I_D \qquad (8)$$

can be obtained for a given device in a wide range of operating junction temperatures T_j, where V_D, V_γ, and R_D are the forward voltage, the threshold voltage, and the dynamic resistance, respectively, under a given forward current I_D. This linear model can be used in order to model a whole lighting fixture—LED driver together with LED lamp—dynamically for a wide range of operating ambient temperatures by setting the expected limit values of forward voltage V_D and dynamic resistance R_D. Table III shows the parameters of the linear model at a junction temperature of 25 °C at 350 mA for the four samples under test.

In this way, a sharp change in the operation temperature, which could even compromise the operation of a LED driver, may be predicted and accounted for. This issue could be critical in lamps with several LEDs in series, since not only the dc

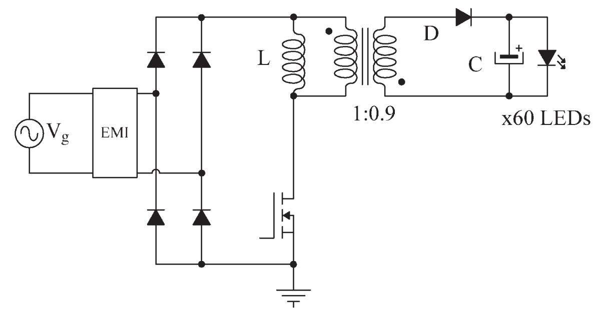

Fig. 13. Typical PFC DCM flyback LED driver.

operation point but also the poles of the transfer function are affected by a shift in the dynamic resistance [10]. In addition, the low-frequency ac ripple would likely be increased out of the specifications with certain output capacitor designs, even leading to unnoticeable but biologically affecting flicker [23].

1. Considerations Over the Output-Current Ripple

As an example, a single-stage current-controlled discontinuous-conduction-mode (DCM) flyback LED driver featuring power factor (PF) correction (PFC) will be considered running a 60-series XLamp XR-E LED lamp at 350 mA, as shown in Fig. 13. In this case, a junction temperature rise from ambient temperature, i.e., 25 °C, corresponding to the start, to a steady-state temperature of 80 °C on this lamp would imply a shift in the dynamic resistance from 57 Ω down to approximately 39 Ω. In other words, the final dynamic resistance would be 65% of the initial value.

The output capacitor has to be sized in order to assure an admissible low-frequency output-current ripple. The averaged current through the output diode, D in Fig. 13, of an offline DCM flyback converter can be expressed as [24]

$$i_D(t) = \frac{V_g^2 d^2}{L f V_o} \sin^2(\omega t) \tag{9}$$

where V_g is the ac supply rms voltage, d is the duty cycle, f is the switching frequency, V_o is the output voltage, L is the inductance of the flyback primary inductor, and ω is the line frequency. This expression is obtained from the processing of power from the rectified line voltage, being characterized by a dc level and an ac component at twice the line frequency. The duty cycle is considered constant to feature PFC while not introducing additional harmonics in the input current. However, this expression can be simplified by substituting the squared sine by its trigonometric equality. Thus

$$i_D(t) = \frac{V_g^2 d^2}{2L f V_o} - \frac{V_g^2 d^2}{2L f V_o} \cos(2\omega t). \tag{10}$$

The first term stands for the dc output current I_o, whereas the second term stands for the double-line-frequency ac component $i_{\mathrm{ac}}(t)$, which will be filtered by the output capacitor C. The main waveforms are depicted in Fig. 14, where the line voltage $V_g(t)$, the average input current $i_{g\_\mathrm{AVG}}(t)$, the actual input current $i_g(t)$, the output diode current $i_D(t)$, and the output average current I_o are highlighted. The averaged output equivalent circuit of the DCM flyback PFC is sketched in Fig. 15.

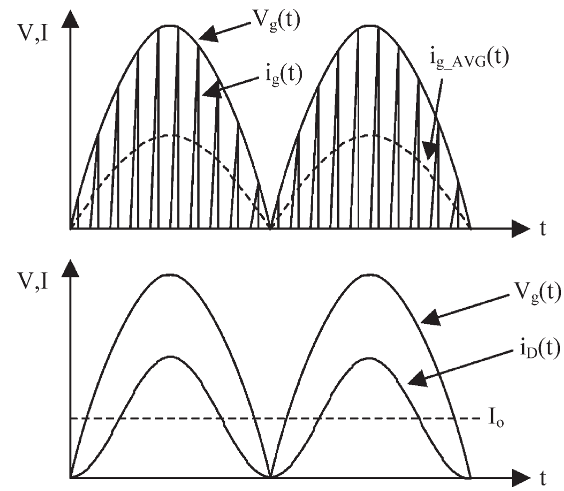

Fig. 14. Ideal main waveforms of the DCM flyback PFC.

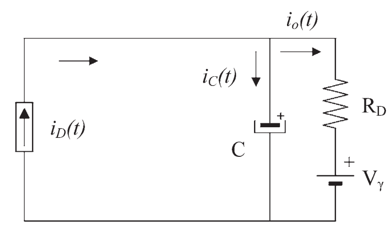

Fig. 15. Averaged output equivalent circuit of the DCM flyback PFC.

This equivalent circuit can be divided into two equivalent subcircuits corresponding to the dc current level, namely, I_o and the ac component $i_{\mathrm{ac}}(t)$. The minimum capacitance for a given output-current ripple can be obtained by accounting for the ac components and using phasor calculus. The ac component through the output diode $\bar{I}_{\mathrm{ac}}$ is the sum of the ac current through the output capacitor $\bar{I}_C$ and the LED load $\bar{I}_o$

$$\bar{I}_{\mathrm{ac}} = \bar{I}_C + \bar{I}_o. \tag{11}$$

First, a ripple factor relating the output-current ac-component amplitude to the dc component can be defined as

$$\delta = \frac{\|\bar{I}_o\|}{I_o}. \tag{12}$$

Second, considering that the output voltage will be determined by the product of the ac output current and the parallel impedance Z_o composed of the output capacitance and the dynamic resistance of the LED load and as it will match the product of the dynamic resistance and the ac output current, the following equality is achieved:

$$Z_o \|\bar{I}_{\mathrm{ac}}\| \angle\phi_{\mathrm{ac}} = R_D \|\bar{I}_o\| \angle\phi_o \tag{13}$$

where ϕ_{ac} and ϕ_o stand for the diode-current and output-current ac-component phase shifts, respectively.

By using phasor calculus, the phase angle between the currents and the complex impedances can be ignored in order to determine the minimum output capacitance for a given ripple factor, considering only the amplitude. Since the purpose of this paper is to check the output-current ripple variation due to the thermal effects on LED lamps and provided that the equivalent series resistance (ESR) of low-ESR capacitors is much lower than the LED lamp dynamic resistance, the capacitor ESR is neglected in this example for the sake of simplicity. Thus, by taking the absolute values in (13), substituting (12) in (13), and rearranging, it could be demonstrated that the minimum capacitance value for a given low-frequency output-current ripple amplitude is given by the following expression:

$$C_{\min} = \sqrt{\frac{1-\delta^2}{4\delta^2\omega^2 R_D^2}} \tag{14}$$

which, as can be seen, depends on the dynamic resistance of the LED lamp, with decreasing minimum capacitance values for increasing dynamic resistance. This suggests that decreasing dynamic resistances will induce higher output-current ripple values.

2. *Considerations Over the Closed-Loop Dynamics*

With regard to the dynamic behavior of the DCM flyback converter, its transfer function can be obtained from the analysis of the dc component of the output equivalent circuit depicted in Fig. 15. For the sake of simplicity, the output capacitor ESR will be neglected. Provided that low ESR values introduce a high-frequency zero and this example illustrates the closed-loop operation of a slow-dynamics LED driver, this simplification could be considered accurate enough in order to predict the most significant effects of thermal variation in the LED lamp. Thus

$$\langle i_D\rangle(t) = \langle i_C\rangle(t) + \langle i_o\rangle(t) \tag{15}$$

where $\langle\ \rangle$ stands for the averaged variables. Developing (15),

$$\frac{\langle v_g\rangle^2\langle d\rangle^2}{2Lf\langle v_O\rangle} = C\frac{d}{dt}\langle v_O\rangle + \langle i_o\rangle. \tag{16}$$

Substituting (8) in (16),

$$\frac{\langle v_g\rangle^2\langle d\rangle^2}{2Lf\left(V_\gamma + R_D\langle i_O\rangle\right)} = R_D C\frac{d}{dt}\langle i_O\rangle + C\frac{d}{dt}\langle V_\gamma\rangle + \langle i_o\rangle. \tag{17}$$

By perturbing and linearizing,

$$R_D C\frac{d}{dt}\langle\widehat{\iota_O}\rangle + \langle\widehat{\iota_O}\rangle = \frac{2V_g^2 d}{2LfV_o}\langle\hat{d}\rangle - \frac{V_g^2 d^2 R_D}{2LfV_o^2}\langle\widehat{\iota_O}\rangle. \tag{18}$$

Rearranging and taking the Laplace transform, the following transfer function is yielded:

$$G(s) = \frac{I_o(s)}{d(s)} = \frac{2\cdot\frac{I_o}{dR_D C}}{s + \frac{V_o + I_o R_D}{V_o R_D C}} \tag{19}$$

where I_o, D, R_D, C, and V_o are the output current, duty cycle, dynamic resistance, output capacitance, and output voltage, respectively. As can be seen, the pole and the static gain of the transfer function depend on the output voltage and the dynamic resistance, which are both temperature-dependent variables.

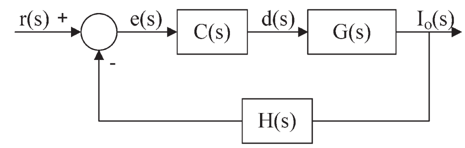

Fig. 16. Closed-loop block diagram of the proposed illustrative example.

The block diagram sketched in Fig. 16 will be considered in this example, where $r(s)$ is the output-current reference, $e(s)$ is the error signal, $C(s)$ is the controller transfer function, $d(s)$ is the control signal applied to the DCM flyback converter $G(s)$, $H(s)$ is the feedback block, and $I_o(s)$ is the output current. Unity feedback is considered for the sake of simplicity.

A proportional–integral compensator will be considered in this illustrative example. The transfer function of this compensator is as follows:

$$C(s) = K_{\mathrm{dc}}\frac{1+zs}{s} \tag{20}$$

where $1/z$ is the compensator-zero corresponding angular frequency and K_{dc} is the compensator static gain.

3. *Example of Dynamic-Resistance Shift Implications*

This example will show a converter designed for a nominal junction temperature of 25 °C, 0.35 duty cycle, 500-μH primary flyback inductor, 57-Ω dynamic resistance, 189-V output voltage, and 180-μF output capacitor for a 30% peak-to-peak output-current ripple target, as extracted from (14).

The compensator will be designed for a 60° phase margin for the nominal junction temperature in order to achieve a fast transient response, and -20-dB gain at 100 Hz in order to prevent the 100-Hz low-frequency ripple from affecting the duty cycle and, thus, the PF. With this constraint, the static gain would be $K_{\mathrm{dc}} = 105.44\ \mathrm{V}^{-1}$, considering a unity-gain PWM modulator, and the zero would be located at 354 rad/s.

Moreover, the cases of 0 °C and 80 °C junction temperatures will be considered in order to analyze the implications of the dynamic-resistance shift on the performance of the converter. Considering these values, the actual output voltages will be 196.8 and 174 V for 0 °C and 80 °C, respectively, and the dynamic-resistance values will be 63 and 39 Ω for those temperatures, respectively. The open-loop Bode diagrams for the three cases are shown in Fig. 17. As can be seen from the picture, the pole of the transfer function would be located at $\omega = 107.8$ rad/s, and its static gain would be $G_{\mathrm{dc}} = 1.78$ A at a junction temperature of 25 °C. However, if the junction temperature rose up to 80 °C, the transfer function pole would be moved to $\omega = 153.63$ rad/s, and the static gain would be shifted to $G_{\mathrm{dc}} = 1.91$ A, as the output voltage would have dropped to 174 V and the dynamic resistance would have been lowered down to 39 Ω. On the other side, if the junction temperature drops down to 0 °C, the static gain would be $G_{\mathrm{dc}} = 1.73$ A, and the pole would be displaced to $\omega = 98.1$ rad/s. This issue might be critical and should be considered in those designs where the feedback control is designed for only one single operating point, since not only the change over time of the closed-loop parameters but also the change in the

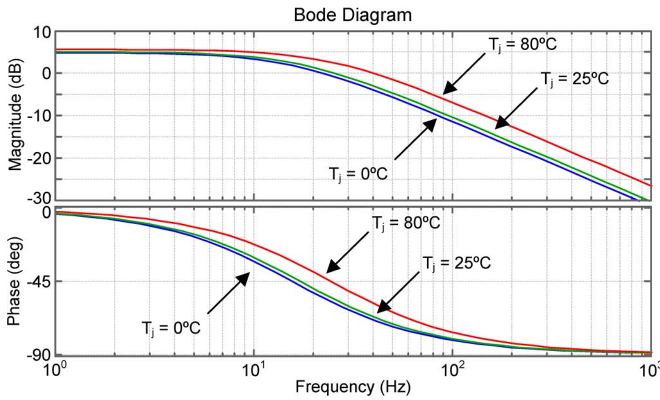

Fig. 17. DCM flyback output-current-to-duty-cycle Bode diagrams for junction temperatures of 0 °C, 25 °C, and 80 °C.

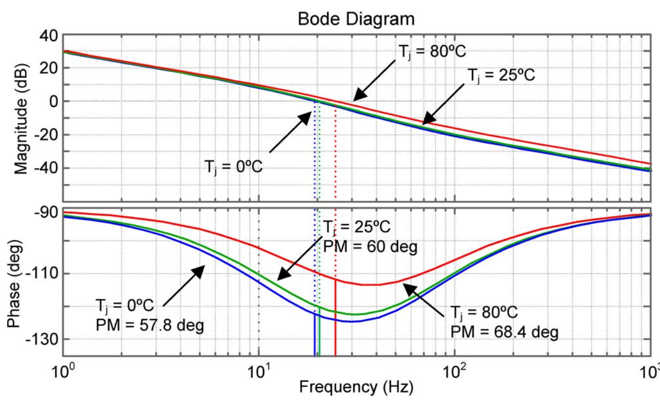

Fig. 18. Loop gain Bode diagrams for junction temperatures of 0 °C, 25 °C, and 80 °C.

thermal operation point could lead to instabilities since the low-frequency poles and the output impedance are affected. Fig. 18 shows the $C(s)G(s)H(s)$ loop gain transfer function with unity feedback for the three junction temperatures considered. As can be noticed, the phase margin is reduced from 68.4° at a junction temperature of 80 °C, down to 57.8° as the temperature decreases to 0 °C.

In order to check the implication of the phase margin shift due to thermal effects, a closed-loop step response under the three junction temperatures considered will be investigated. The results are shown in Fig. 19. As can be seen, as the phase margin is boosted, corresponding to a lower dynamic resistance due to a rise in junction temperature, the closed-loop system becomes less oscillatory and features a lower overshoot, with the settling time being also reduced. On the contrary, as the junction temperature is decreased inducing a higher dynamic-resistance value, which corresponds to a lower phase margin, the step response becomes more oscillatory, with higher overshoot and larger settling times.

Regarding the implications of the junction temperature shift on the output-current ripple, the following value for the current ripple is achieved after rearranging (14):

$$\delta = \frac{1}{\sqrt{1 + 4\omega^2 C^2 R_D^2}}. \tag{21}$$

With a 180-μF output capacitor, the theoretically calculated low-frequency output-current peak-to-peak ripple is set at 31% of the dc value at nominal operation, whereas the ripple is raised up to 44% at a junction temperature of 80 °C. On the contrary, as the dynamic resistance is increased as long as the junction temperature is decreased, the ripple will be lower at 0 °C, being 28% over the dc current. Fig. 20 shows the output-current ripple under closed-loop operation at the three junction temperatures. As can be seen from this figure, the peak-to-peak output-current ripple is set to approximately 100 mA at the nominal junction temperature, which represents around a 30% peak-to-peak current ripple. However, the output-current ripple swings from less than 100 mA at 0 °C to more than 140 mA at 80 °C, confirming the theoretical calculations.

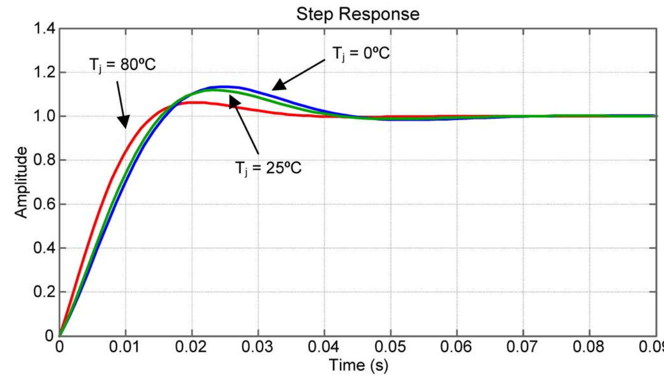

Fig. 19. Step response for junction temperatures of 0 °C, 25 °C, and 80 °C.

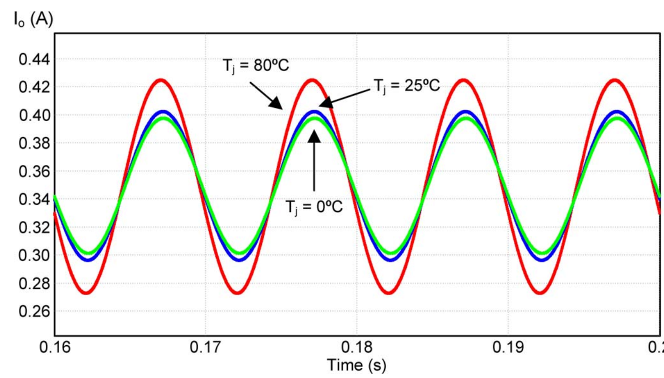

Fig. 20. Closed-loop output-current ripple simulation for junction temperatures of 0 °C, 25 °C, and 80 °C.

As can be extracted from this illustrative example, the dynamic-resistance shift caused by the junction temperature change induces two effects on the closed-loop operation of a LED driver. On the one hand, decreasing dynamic resistances as a consequence of a junction temperature change would lead to increasing low-frequency output-current ripples. On the other hand, increasing dynamic resistances as a consequence of a junction temperature shift would lead to a more underdamped response.

V. Conclusion

The effects of the junction temperature on the electrical behavior of LEDs have been tested in this work. First, four commercial GaN-based phosphor-converted white LEDs from four of the main manufacturers have been characterized using the voltage–temperature method in order to determine the junction temperature accurately in subsequent tests. The

TABLE IV
GOLDEN DRAGON PLUS V_D–T_J DEPENDENCE (POLYNOMIAL FIT)

I_D (mA)	$V_D=a\cdot T_j^2+b\cdot T_j+c$ (V)			R^2
	a (x10^{-6})	b (x10^{-3})	c	
100	2.301	–2.726	2.996	0.9989
200	3.825	–3.020	3.106	0.9989
300	3.450	–3.015	3.187	0.9988
350	4.082	–3.107	3.225	0.9993
400	4.544	–3.121	3.257	0.9998
500	3.974	–3.086	3.322	0.9994
600	3.980	–3.035	3.379	0.9992
700	4.584	–3.073	3.435	0.9991
800	5.185	–3.132	3.488	0.9980
900	6.309	–3.184	3.538	0.9992
1000	6.155	–3.248	3.591	0.9981

experimental results have shown that all the devices tested featured a quadratic dependence of the forward voltage drop as opposed to the junction temperature for a given injected current. However, two different behaviors were checked: Whereas the Golden Dragon Plus and the Luxeon K2 with TFFC performed more linearly, the Z-Power P4 and the XLamp XR-E featured a more pronounced quadratic dependence. These results refute the linear dependence generally assumed for a wide temperature range [16]. Nevertheless, the experimental results agree with the forward-voltage–junction-temperature relation provided by some manufacturers [13]–[15].

Second, the influence of the junction temperature on the LED electrical behavior has been tested and checked. In this way, a small-signal analysis has been performed at three injection current levels and variable junction temperature in order to determine the junction temperature and the dynamic resistance. The experimental results confirm that the dynamic resistance features a shift due to changes in the junction temperature. However, the change in the dynamic resistance has shown different behaviors, as the dynamic resistance underwent a drop as the junction temperature was raised only in three of the samples tested. In the remaining device, the actual change in the dynamic resistance did happen as opposed, since this parameter was increased as the junction temperature rose up at the highest test currents but remained fairly constant at the lowest test current for the whole temperature range.

Furthermore, the experimental results achieved show that a sharp change in the operation temperature could even compromise the operation of both the LED fixtures due to a large shift in the dynamic resistance of several series-connected LEDs. Depending on the commercial device and the lamp design, this may cause a more underdamped response of the output current as the junction temperature decreases, as well as an increase in the output-current ripple, showing that both effects should be taken into account, particularly when the long-term performance of the lighting fixture under a wide range of operation temperatures is eventually expected.

Finally, with the experimental procedures described in this paper, a linear model can be obtained for a wide range of operating junction temperatures, providing the designer with the actual linear model for each injection current at each junction temperature, so either the power stage or the feedback loop could be optimized for a wide range of operation conditions of the final lighting product or fixture.

TABLE V
LUXEON K2 WITH TFFC V_D–T_J DEPENDENCE (POLYNOMIAL FIT)

I_D (mA)	$V_D=a\cdot T_j^2+b\cdot T_j+c$ (V)			R^2
	a (x10^{-6})	b (x10^{-3})	c	
100	4.021	–3.766	3.216	0.9986
200	4.694	–4.014	3.380	0.9987
300	4.101	–4.098	3.505	0.9990
350	4.855	–4.269	3.560	0.9988
400	4.102	–4.266	3.643	0.9991
500	3.456	–4.285	3.703	0.9990
600	4.898	–4.552	3.794	0.9986
700	2.608	-4.355	3.864	0.9993
800	4.359	–4.556	3.938	0.9985
900	3.254	–4.598	4.018	0.9992
1000	3.086	–4.643	4.089	0.9982

TABLE VI
Z-POWER P4 V_D–T_J DEPENDENCE (POLYNOMIAL FIT)

I_D (mA)	$V_D=a\cdot T_j^2+b\cdot T_j+c$ (V)			R^2
	a (x10^{-5})	b (x10^{-3})	c	
100	1.041	–3.864	3.015	0.9985
200	1.252	–4.640	3.200	0.9990
300	1.579	–5.340	3.341	0.9986
350	1.696	–5.595	3.403	0.9986
400	1.916	–6.037	3.468	0.9985
500	2.014	–6.377	3.576	0.9967
600	2.244	–6.851	3.680	0.9989
700	2.310	–7.150	3.777	0.9989
800	2.447	–7.525	3.871	0.9993
900	2.492	–7.835	3.964	0.9991
1000	2.468	–8.075	4.057	0.9991

TABLE VII
XLAMP XR-E V_D–T_J DEPENDENCE (POLYNOMIAL FIT)

I_D (mA)	$V_D=a\cdot T_j^2+b\cdot T_j+c$ (V)			R^2
	a (x10^{-5})	b (x10^{-3})	c	
100	1.360	–3.838	2.914	0.9959
200	2.088	–5.399	3.076	0.9964
300	2.615	–6.640	3.213	0.9971
350	2.770	–7.082	3.274	0.9980
400	2.866	–7.445	3.331	0.9986
500	2.966	–8.070	3.440	0.9986
600	3.116	–8.613	3.524	0.9992
700	3.180	–9.021	3.623	0.9996
800	2.966	–9.120	3.705	0.9997
900	2.926	–9.365	3.785	0.9990
1000	2.814	–9.539	3.864	0.9993

APPENDIX

The V_D–T_J dependences of the Golden Dragon Plus, Luxeon K2, Z-Power P4, and XLamp XR-E devices are shown in Tables IV–VII, respectively.

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